

Activity - 1

① Medical image Enhancement

Scenario:- An X-ray image shows bones clearly, but darker tissues have poor visibility.

1. power-law (Gamma) Transformation with $\gamma < 1$.

2. * It increases the brightness of dark regions.

* It enhances soft tissues without significantly affecting the already visible bones.

3. * Dark tissues become brighter and easy to observe.

* Soft tissue become clearer

② Night - Time Surveillance

Scenario:- A CCTV image is very dark because most pixel values lie b/w 0 and 40.

1. * Log Transformation.

2. * Log transformation expands

low-intensity (dark) pixel values.
* It reveals hidden details in dark region.

3. * The parking lot becomes brighter
* vehicles and people become easier to identify.

* Dark areas reveal more detail.

③ Fingerprint Recognition.

Scenario: - A fingerprint image has low contrast b/w ridges and valleys.

1. * contrast stretching

(piecewise linear transformation)

2. * It increases the contrast b/w ridges and valleys

* Fine fingerprint become more distinct.

* It improves feature extraction and matching accuracy.

3. * log transformation mainly dark brighter region.

* It doesn't effectively increase the contrast b/w ridges and valleys.

* Fingerprint details may appear brighter but remain less distinguishable than with contrast stretching.

* Medical X-ray $\rightarrow$ power-law (gamma $\neq 1$)

* Night-time CCTV $\rightarrow$ log transformation.

* Fingerprint Recognition $\rightarrow$ contrast stretching